The Impact of Economic Policy Uncertainty on Bilateral Trade Flows in Differentiated Goods Sectors

by

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January 2026

1 Introduction

Economic policy uncertainty (*EPU*) has emerged as a critical determinant with a direct effect on bilateral trade flows, as they are impacted by exporters' and importers' economic policy uncertainty (Borojo et al., 2021). More recent evidence highlights substantial differences across different sectors and time horizons. Focusing on differentiated goods sectors, Nana et al., 2025 show that economic policy uncertainty has heterogeneous effects for differentiated goods. Understanding these differentiated sectoral responses is therefore essential to analyse bilateral trade flows in times of geopolitical and economic turbulence.

This paper investigates how economic policy uncertainty originating in both exporting and importing countries affects bilateral trade flows across three disaggregated sectors: capital goods, intermediate goods, and consumption goods. Therefore, we focus on a dynamic lagged dimension examining immediate effects alongside lagged impacts at two- or four-year horizons.

The gravity model of trade has been developed as the baseline framework for analysing bilateral trade flows, with its microfoundations established in the structural gravity approach of Anderson & Wincoop, 2003. A central challenge in applying this framework is related to the treatment of multilateral resistance terms (MRT), aggregates of bilateral trade costs that capture effects in a general equilibrium setting. Baldwin & Taglioni, 2006 therefore warn that when MRT are not properly accounted for, estimates suffer from omitted variable bias and fail to capture how trade costs with all partners affect bilateral relationships. This results in a challenge when analysing the effects of country-specific time-varying variables such as economic policy uncertainty, which are perfectly collinear with the exporter-time and importer-time fixed effects that are typically proxies for multilateral resistance terms.

Methodologically, we address this challenge by employing two estimation strategies, each with distinct advantages. First, we implement a one-step PPML à la Silva & Tenreyro, 2006 that includes proxies for multilateral resistances. Second, we adopt a two-stage structural gravity procedure formalised by Head & Mayer, 2014 and recently applied by Matzner et al., 2023, observing uncertainty effects through bilateral and multinational channels where exporter uncertainty increases outward multilateral resistance (reducing trade by 0.76%) , while importer uncertainty has the opposite effect. In a first stage, we therefore estimate the gravity equation with exporter-time, importer-time, and pair fixed effects to recover theoretically consistent MRT indexes, as advocated by Yotov & Larch, 2016. This underlines the property demonstrated by Fally, 2015 that PPML estimation with three-way fixed effects automatically satisfies the adding-up constraints inherent

in structural gravity models. In a second stage, we regress the recovered fixed effects on our lagged EPU measures, GDP, population and other time-dependent controls.

The remainder of the paper is structured as follows. Section 2 outlines the theoretical framework and estimation strategies. Section 3 presents the results from both the one-step and two-stage approaches. Section 4 concludes.

2 Empirical Strategy

In order to assess whether and how economic policy uncertainty affects bilateral trade, our empirical strategy is founded on two parts. In the first part, we analyse the problem following a *naive* approach, in which the gravity equation is estimated within a one-step framework, without the inclusion of fixed effects. In the second part, we address the problem using a two-stage estimation strategy, in order to be more consistent with state-of-the-art econometric recommendations for the estimation of *structural* gravity models.

2.1 The One-Step Method: A *Naive* Approach

Within the one-step approach, the gravity model specification is:

$$X_{ij,t} = \exp\left(\mathbf{Z}'_{ij}\boldsymbol{\beta} + \mathbf{B}'_{ij,t}\boldsymbol{\rho} + \mathbf{W}'_{i,t}\boldsymbol{\theta} + \mathbf{V}'_{j,t}\boldsymbol{\kappa} + \sum_{\ell\in\{0,2,4\}} \gamma_\ell \ln \mathrm{EPU}_{i,t-\ell} + \sum_{\ell\in\{0,2,4\}} \delta_\ell \ln \mathrm{EPU}_{j,t-\ell}\right)\cdot\varepsilon_{ij,t}, \qquad i \neq j. \tag{1}$$

where $X_{ij,t}$ denotes exports from country i to country j at time t. The EPU index is introduced on both the exporter and importer sides, considering also lagged values at two and four periods. The set of covariates, expressed in logarithmic form when quantitative, is organised as follows. The vector $\mathbf{Z}_{ij}$ collects time-invariant bilateral characteristics, including distance, contiguity, common language, and colonial ties. The vector $\mathbf{B}_{ij,t}$ includes time-varying bilateral controls, namely the presence of a regional trade agreement. Finally, $\mathbf{W}_{i,t}$ and $\mathbf{V}_{j,t}$ gather exporter- and importer-time–specific variables, respectively, namely nominal GDP, population, WTO membership, and the remoteness index.

All variables are described in detail in Table 1. The choice of covariates follows standard theoretical considerations. Dyadic variables contained in $\mathbf{Z}_{ij}$ and $\mathbf{B}_{ij,t}$ are interpreted as proxies for bilateral trade costs, as in standard empirical gravity models. Exporter- and importer-specific time-varying variables are included to capture exporter's production, importer's expenditure and inward and outward multilateral resistances. GDP is used as the standard proxy for economic size, while population helps to better capture expenditure capacity, since, conditional on GDP, it serves as a proxy for a country's income level. Incorporating a measure that more accurately reflects expenditure capacity is theoretically consistent with the standard gravity equation, particularly on the importer side.

Finally, WTO membership and the outward and inward remoteness indexes are included as proxies for multilateral resistance terms, capturing a country's relative trade openness and geographic position. Importantly, the remoteness index in this specification is constructed using only the 21 countries for which EPU index data are available. Although this approach is not fully coherent with the structural gravity framework since multilateral resistance terms are defined over the entire global trading network, it remains internally consistent from an econometric perspective.

It is important to acknowledge that specification (1) is subject to two limitations. First, it is limited by not relying on a theoretically consistent fixed-effects approach to estimate multilateral resistance terms, weakening the consistency of our estimates. More specifically, solely relying on proxies provides a poor approximation of multilateral resistance, as pointed out by Head & Mayer, 2014. Secondly, this specification can only be estimated on a small set of economies where the risk of bias is substantial. This reflects not only standard econometric sample selection concerns, but also a gravity-specific issue: since trade flows are embedded in a global network, restricting the analysis to a limited subset of countries and bilateral relationships can generate severe distortions.

Despite these limitations, specification (1) provides a valuable robustness check, which lies in the advantage of richer data availability: it allows us to assess the effect of economic policy uncertainty, using a substantially larger number of observations than in the subsequent two-stage approach.

2.2 The Two-Stage Method: A *Structural* Approach

The main limitation of the approach presented in Section 2.1 is that, since the EPU index is country-specific and time-varying, identifying the associated coefficients requires excluding exporter–time and importer–time fixed effects from the specification. However, the state-of-the-art econometric literature strongly recommends the use of fixed effects when estimating gravity models with panel data (Yotov & Larch, 2016). In particular, exporter–time and importer–time fixed effects are crucial to properly control for multilateral resistance terms and other unobserved country-specific shocks. Their inclusion allows the estimation of a gravity specification that is fully consistent with the structural gravity framework.

We therefore adopt a two-stage approach (Head & Mayer, 2014). In the first stage, we estimate a gravity model including country-pair fixed effects as well as exporter–time and importer–time fixed effects. In the second stage, the estimated exporter–time and importer–time fixed effects are regressed on the exporter and importer EPU indexes, respectively, along with additional country-specific time-varying covariates.

Notably, the two-stage approach also addresses an additional issue discussed in Section 2.1, namely the network nature of the structural gravity model. In particular, as will be discussed, the first stage, which is estimated using a fixed-effects approach, allows the gravity equation to be fitted on the full set of country-time observations, rather than on a restricted subsample as in the one-step approach. As a result, multilateral resistance terms are structurally estimated in the first stage. This preserves the interpretation of exporter–time and importer–time fixed effects as theory-consistent measures of outward and inward multilateral resistance within the global trading network, thereby properly accounting for the structural nature of the gravity model itself. What remains, however, is the presence of sample selection concerns in the second-stage regression. Nevertheless, at this point the multilateral resistance terms have already been consistently estimated in the first stage and fully account for the global network structure of trade.

Even though the two-stage approach is more coherent with the theoretical framework of structural gravity estimation, it also entails a drawback that becomes particularly severe in our setting, namely the limited number of observations. Specifically, the fixed effects recovered in the first step are defined at the country–time level, implying a country-by-year matrix that already substantially reduces the effective sample size, a limitation that is further exacerbated by the restricted number of countries with availability of the EPU index.

2.2.1 The First Stage

The first stage is based on the estimation of a canonical structural gravity specification in which pair fixed effects, exporter–time fixed effects, and importer–time fixed effects are included, together with $RTA_{ij,t}$ as a proxy for time-varying bilateral trade cost components. The specification is given by

$$X_{ij,t} = \exp\Big(\lambda_{i,t} + \psi_{j,t} + \mu_{ij} + \beta_1 \,\mathrm{RTA}_{ij,t}\Big) \cdot \varepsilon_{ij,t}, \qquad i \neq j. \tag{2}$$

Here, $\lambda_{i,t}$ denotes the exporter–time fixed effect, $\psi_{j,t}$ the importer–time fixed effect, and μ_{ij} the bilateral (pair) fixed effect; $RTA_{ij,t}$ is a dummy that takes value one if countries i and j are members of a regional trade agreement at time t, and zero otherwise.

Relevantly, the inclusion of country-pair fixed effects and country–time fixed effects, as proposed in specification (2), absorbs a large set of observable covariates. More precisely, pair fixed effects capture all time-invariant dyadic characteristics. Exporter–time and importer–time fixed effects instead absorb all monadic, time-varying country-specific variables, including the EPU index. Consequently, these variables cannot be separately identified in the first-stage gravity estimation and have to be analysed in a second-stage framework based on the estimated fixed effects.

2.2.2 The Second Stage

In the second stage, the exporter-time and importer-time fixed effects obtained from specification (2), denoted respectively by $\widehat{\lambda}_{i,t}$ and $\widehat{\psi}_{j,t}$, are regressed on a set of country–time covariates according to the following reduced-form specifications:

$$\widehat{\lambda}_{i,t} = \mathbf{X}'_{i,t}\boldsymbol{\beta} \;+\; \sum_{\ell \in \{0,2,4\}} \gamma_\ell \ln \mathrm{EPU}_{i,t-\ell} \;+\; \delta_t \;+\; u_{i,t}, \tag{3}$$

$$\widehat{\psi}_{j,t} = \mathbf{X}'_{j,t}\boldsymbol{\gamma} \; + \sum_{\ell \in \{0,2,4\}} \delta_\ell \ln \mathrm{EPU}_{j,t-\ell} \; + \; \delta_t \; + \; v_{j,t}, \tag{4}$$

The EPU index is included on both the exporter and importer sides, allowing for contemporaneous effects as well as delayed responses through lags of two and four periods. The vectors $\mathbf{X}_{i,t}$ and $\mathbf{X}_{j,t}$ collect country–time covariates, such as GDP, WTO membership, population, and the remoteness index, which is now defined over the full set of countries. The term δ_t denotes a full set of year fixed effects, absorbing global shocks and common time trends affecting all countries symmetrically. Details of the variables are defined and summarised in Table 1.

The selection of the explanatory variables is theoretically motivated as follows. Econometrically, the exporter fixed effects $\widehat{\lambda}_{i,t}$ and the importer fixed effects $\widehat{\psi}_{j,t}$ absorb two distinct components of structural gravity. The first component relates to country size, and more precisely to production on the exporter side and expenditure on the importer side. These elements are proxied by GDP and population, in line with the theoretical discussion in Section 2.1. The second component corresponds to multilateral resistance terms, which capture a country's overall ease of access to foreign markets or exposure to foreign competition. Multilateral resistance is therefore linked to measures of remoteness and institutional integration, such as WTO membership, as well as to aggregate uncertainty conditions, which are proxied by the EPU index and its lagged values.

Importantly, all quantitative covariates are entered in logarithmic form. This choice is consistent with the specification in equation (2), where the fixed effects were estimated as log-components of the structural gravity equilibrium.

2.3 Data

Considering that the analysis relies on two different approaches, we construct two distinct but closely related datasets, `Dataset_Filtered.dta` and `Dataset_Full.dta`, each tailored to the requirements of the respective empirical strategy. Both datasets are obtained by merging the bilateral trade database (`Gravity_M2IT.dta`), provided for teaching purposes by Toubal, 2025, with the EPU index dataset (`EPU.dta`) developed by Baker et al., 2016.[1]

Toubal, 2025 comprises 208 economies covering a period from 2000 to 2019 including information on countries' bilateral trade flows, specifications on category of goods traded (e.g. consumption, intermediate, capital goods), the amount of traded products, explanatory variables like distance measures (population weighted, capital, most populated cities distances), GDP and several more cultural measures (e.g. common language, civil law, legal origin, former colonial relationships, etc.) resulting in 553,377 single row observations.

We retrieve information of economic policy uncertainty as developed in Baker et al., 2016, serving as the time-varying proxy for institutional and policy-related trade-costs in the form of all summarised country- and GDP weighted indices for a full time-period of 1997 until 2025.

The two main datasets differ in their treatment of EPU availability and in the time dimension considered. `Dataset_Full.dta` preserves the complete set of bilateral country relationships contained in `Gravity_M2IT.dta`, including observations for which EPU data are missing for either the exporter or the importer. In contrast, `Dataset_Filtered.dta` restricts the sample to bilateral observations for which EPU data are available on both sides of the trade pair and further applies a time filter to selected years.

Details of the two datasets are reported in Table 2, which highlights the risk of sample selection bias: economies included in `Dataset_Filtered.dta` are, on average, substantially larger in terms of economic size, population, and trade volumes. Considering the construction of the datasets, it follows directly that `Dataset_Filtered.dta` is used to estimate specification (1), while `Dataset_Full.dta` is employed for specification (2).

3 Results

3.1 The One-Step Approach

Estimates of specification (1) are obtained through a Poisson Pseudo–Maximum Likelihood (PPML) estimator to address issues of heteroskedasticity, as emphasised by Silva & Tenreyro, 2006. Consistent with standard gravity literature practice, standard errors are clustered at the country-pair

[1] The EPU data are obtained from http://policyuncertainty.com/all_country_data.html.

level to account for potential correlation of the error term within bilateral pairs. Finally, to include adjustment dynamics in trade flows, the panel is constructed at discrete time intervals, with observations taken at three-year horizons, namely $t \in (2007, 2010, 2013, 2016, 2019)$. Table 3 reports the results of the naive gravity specification. Column (1) presents the estimates for total bilateral trade, while Columns (2)–(4) report the results for disaggregated trade flows in capital, intermediate, and consumption goods, respectively.

We can confirm that standard gravity forces behave as expected: distance is consistently negative and significant across all sectors, while GDP is strongly positive, with elasticities close to one. Bilateral controls such as contiguity and regional trade agreements are also positive and significant across all sectors, confirming their role in reducing trade costs.

For economic policy uncertainty, the results are broadly close to hypothesised mechanisms. On the exporter side (EPU_i), present uncertainty is statistically insignificant, while significant negative effects emerge with a lag, consistent with Yotov & Larch, 2016 that trade adjustments are not instantaneous. More precisely, aggregate trade, intermediate inputs, and capital goods display a statistically significant negative coefficient at lag $_{t-4}$, while for capital goods a strong negative effect is already evident at lag $_{t-2}$. Instead, consumption goods respond at lag $_{t-2}$. On the importer side (EPU_j), the results are less severe, though we observe a significant negative effect at $_{t-2}$ for aggregate trade and intermediate goods. From these results, we can draw preliminary conclusions. Firstly, our results support the view that supply-side uncertainty has a strong and statistically significant negative impact on future exports, while effects on the importer side are weaker and less clearly identifiable. Secondly, the impact of uncertainty is most observed in lagged periods, with present effects being generally insignificant. Thirdly, there are heterogeneous effects between sectors, with capital goods being strongly sensitive to exporter-side uncertainty, observing larger coefficients and significant effects already at shorter lags, which indicates a faster response. Further, consumption goods react more quickly at lag $_{t-2}$ on the exporter side, but the effect fades out thereafter.

3.2 The Two-Stage Approach

3.2.1 First Stage

The estimation of specification (2) is carried out using the PPML estimator, with standard errors clustered at the country-pair level to account for potential intra-cluster correlation within trading pairs. The results are reported in Table 4.

Importantly, as mentioned, given the structure of specification (2), the full set of country pairs is retained, including those for which the EPU indices are not observed, thereby preserving a structural approach in which exporter-time and importer-time fixed effects capture trade relationships in a general equilibrium setting within the global trade network.

Notably, since both country-specific time-varying fixed effects and pair fixed effects are included in the specification, redundancies are required to avoid multicollinearity: one redundant exporter fixed effect and twenty importer fixed effects, one for each year.

The estimated coefficients on the regional trade agreement (RTA) indicator are positive and highly statistically significant for all trade aggregates, except for capital goods. However, it is worth noting that the magnitude of the coefficients is substantially smaller than that typically found in the literature; this may be due to the absorbing effects induced by the fixed effects on the RTA variable. Finally, the pseudo R^2 is very high in all specifications, which is consistent with the inclusion of rich fixed-effect structures that absorb a large share of the variation in bilateral trade flows.

3.2.2 Second Stage

Specifications (3) and (4) are estimated using the ordinary least squares (OLS) method, with standard errors clustered at the country level to account for residual correlation within countries. The corresponding results are reported in Tables 5 and 6, respectively.

Importantly, before turning to any analysis of the results, it is worth mentioning an important, yet subtle, element regarding the second stage. Since, in the first stage, the fixed effects are inferred as part of the structural equilibrium of the gravity model, the regression applied in the second stage should not be interpreted as assessing how these variables affect trade flows directly, but rather as evaluating how they affect the fixed effects and, more importantly, as will be shown, the

multilateral resistance terms. That said, any analysis of how the multilateral resistance terms are affected must be conducted within the framework of structural gravity theory, which is inherently based on indirect implications for bilateral trade.

When fixed effects are included in the first stage, both outward and inward multilateral resistance terms are absorbed, consistently with the canonical structural gravity framework, as well as other time-varying country-specific variables, most notably GDP and population. Importantly, since multilateral resistance terms enter the fixed effects with a negative sign, any variable that is interpreted as affecting multilateral resistance – such as EPU in our specific case – will influence the multilateral resistance terms in the opposite direction to that implied by the fixed-effect regression coefficients.

Primarily, the regressions corresponding to specifications (3) and (4) display a strong and positive coefficient on GDP, which is consistent with its theoretical interpretation as a proxy for country size and, therefore, with its positive association with the fixed effects.

Looking at economic policy uncertainty, the second-stage results confirm the patterns identified in specification (1). On the exporter side, the EPU index exhibits a negative and statistically significant effect only at $_{t-2}$ and $_{t-4}$, while the contemporaneous coefficient is not statistically significant. More precisely, consumption goods display the fastest adjustment, with a statistically significant coefficient already at $_{t-2}$, whereas aggregate trade, intermediate goods, and capital goods reach statistical significance only at the four-year lag, indicating a less immediate response, with capital goods exhibiting the largest negative coefficient. On the importer side, again in line with previous findings, statistical significance is less marked, and the relevant results are mainly observed at $_{t-2}$, with capital goods failing to display a statistically significant coefficient at any lag. Finally, when analysing the results related to this second stage, it is worth noting that the second-stage analysis relies on only 273 observations, which is an obvious consequence of considering exporter- and importer-time–varying effects for only 21 economies over 13 years. From this, two main drawbacks arise: (a) the risk of sample-selection bias in the estimation of the coefficients, and (b) a sample size that is very limited relative to standard gravity estimations based on bilateral trade flows. While point (a) cannot be addressed at this stage given the limited coverage of the EPU index, point (b) does not appear to have been particularly binding in preventing the emergence of meaningful and economically interpretable results, thereby suggesting a strong relationship between economic policy uncertainty and multilateral resistance terms.

4 Conclusion

This paper analysed the impact of economic policy uncertainty on bilateral trade flows across differentiated sectors using a structural gravity framework. The analysis compared a one-step estimation approach with a theoretically consistent two-stage approach that took multilateral resistance terms into account.

Our empirical results showed that economic policy uncertainty resulted in a negative effect on trade, with delayed impacts observable after two- and four-year lags rather than an instantaneous response. On the exporter side, uncertainty strongly reduced future exports across most sectors, with stronger effects for capital goods at the two-year lag and for total trade and intermediate goods at the four-year lag. For importers, uncertainty effects were less prominent but still observable, particularly for intermediate goods at the two-year lag. The heterogeneity between sectors in the analysis underscored that capital, intermediate, and consumption goods responded differently to uncertainty shocks. Capital goods exports showed the strongest sensitivity to exporter uncertainty, while intermediate goods revealed vulnerable exposure to both exporter and importer uncertainty.

Methodologically, our paper highlighted a critical trade-off. The two-stage approach came with theoretically consistent MRTs recovered from the full trade network, avoiding omitted variable bias. However, this resulted in a strongly reduced sample size of 260 country-year observations, due to the limited availability of EPU indices for other countries included in the raw datasets. Therefore, the one-step approach, even if theoretically less robust, served as an interesting benchmark by using a much larger sample of 2,100 observations to validate the structural relationship.

Our main limitation came from the restricted availability of EPU data, which was limited to 21 economies over a shortened time span in our two-stage analysis. Despite this, the results provided reproducible findings for trade policy analysis, emphasising that the adverse trade effects of policy uncertainty were not only sector-specific, but also persistent, often revealed only after years of the initial shock.

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Datasets

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A Appendix

Table 1: Variables description

Variable	Description
$X_{ij,t}$	Exports from country i to country j at time t
$GDP_{i,t}$	Gross domestic product of exporter i at time t (current US$)
$GDP_{j,t}$	Gross domestic product of importer j at time t (current US$)
$POP_{i,t}$	Population of exporter i at time t
$POP_{j,t}$	Population of importer j at time t
$Dist_{ij}$	Weighted bilateral distance between the most populated cities of i and j
$RTA_{ij,t}$	Dummy equal to one if a regional trade agreement between i and j is in force at time t
$EPU_{i,t}$	Economic Policy Uncertainty index of exporter i at time t
$EPU_{j,t}$	Economic Policy Uncertainty index of importer j at time t
$EPU_{i,t-k}$	Lagged Economic Policy Uncertainty of exporter i ($k = 2, 4$)
$EPU_{j,t-k}$	Lagged Economic Policy Uncertainty of importer j ($k = 2, 4$)
$WTO_{i,t}$	Dummy equal to one if exporter i is a WTO member at time t
$WTO_{j,t}$	Dummy equal to one if importer j is a WTO member at time t
$Contig_{ij}$	Dummy equal to one if exporter i and importer j share a common border
$ComCol_{ij}$	Dummy equal to one if exporter i and importer j share a common colonial relationship after 1945
$ComLang_{ij}$	Dummy equal to one if exporter i and importer j share a common official language
$Remoteness_{i,t}$	Outward remoteness index of exporter i at time t
$Remoteness_{j,t}$	Inward remoteness index of importer j at time t

Table 2: Summary statistics of the datasets

	`Dataset_Full.dta`	`Dataset_Filtered.dta`
Observations	553,377	2,520
Countries (exporters / importers)	208 / 208	21 / 21[b]
Time span	2000–2019	Selected years[a]
Trade (mean)	470,866	13,700,000
Nominal GDP (mean)	4.45×10^{11}	2.52×10^{12}
Population (mean)	4.49×10^{7}	1.98×10^{8}
WTO membership (mean)	0.82	0.98
EPU (mean)	131.67	131.67
All countries retained	Yes	No
EPU available for all observations	No	Yes
Designed for specification	Two-stage	One-stage

Notes: Means are computed at the bilateral level. Exporter and importer values are theoretically symmetric and therefore reported in condensed form. [a] Selected years are 2004, 2007, 2010, 2013, 2016, and 2019. [b] The filtered dataset includes Australia, Brazil, Canada, Chile, China, France, Germany, Greece, India, Ireland, Italy, Japan, Mexico, Pakistan, Republic of Korea, Russian Federation, Singapore, Spain, Sweden, United States, and United Kingdom.

Table 3: Estimation of specification (1)

	(1) Trade	(2) Capital	(3) Intermediate	(4) Consumption
$\ln GDP_{i,t}$	0.875***	1.077***	0.885***	0.644***
	(0.052)	(0.076)	(0.053)	(0.060)
$\ln GDP_{j,t}$	1.025***	1.077***	0.892***	1.290***
	(0.079)	(0.120)	(0.062)	(0.130)
$\ln Dist_{ij}$	-0.701***	-0.609***	-0.772***	-0.681***
	(0.053)	(0.084)	(0.056)	(0.071)
$\ln EPU_{i,t}$	-0.054	-0.022	-0.108	0.084
	(0.069)	(0.097)	(0.073)	(0.070)
$\ln EPU_{i,t-2}$	-0.093	-0.402***	0.062	-0.197***
	(0.059)	(0.086)	(0.073)	(0.060)
$\ln EPU_{i,t-4}$	-0.202***	-0.320***	-0.190***	-0.124
	(0.061)	(0.094)	(0.069)	(0.090)
$\ln EPU_{j,t}$	-0.110	-0.108	-0.105	-0.134
	(0.086)	(0.132)	(0.079)	(0.100)
$\ln EPU_{j,t-2}$	-0.126**	-0.011	-0.205***	0.029
	(0.058)	(0.101)	(0.060)	(0.070)
$\ln EPU_{j,t-4}$	-0.020	0.010	-0.037	-0.028
	(0.059)	(0.092)	(0.058)	(0.095)
$RTA_{ij,t}$	0.363***	0.375***	0.350***	0.247**
	(0.101)	(0.135)	(0.111)	(0.120)
$\ln POP_{i,t}$	-0.009	0.156	-0.151**	0.299***
	(0.073)	(0.099)	(0.063)	(0.084)
$\ln POP_{j,t}$	-0.151***	-0.203**	0.013	-0.547***
	(0.055)	(0.080)	(0.049)	(0.077)
$WTO_{i,t}$	0.080	2.520***	-0.400*	0.335
	(0.212)	(0.337)	(0.210)	(0.262)
$WTO_{j,t}$	-0.027	-0.285	0.491*	-0.543***
	(0.239)	(0.297)	(0.294)	(0.199)
$Contig_{ij}$	0.386***	0.432**	0.381***	0.460***
	(0.115)	(0.194)	(0.119)	(0.141)
$ComCol_{ij}$	-0.163	0.232	-0.713	0.247
	(0.582)	(0.756)	(0.538)	(0.386)
$Colony_{ij}$	0.148	0.016	0.216**	0.169
	(0.096)	(0.145)	(0.095)	(0.132)
$Remoteness_{i,t}$	0.421	-0.907***	1.088***	-0.214
	(0.272)	(0.346)	(0.281)	(0.314)
$Remoteness_{j,t}$	0.799***	1.076***	0.626***	0.807***
	(0.174)	(0.236)	(0.188)	(0.190)
Observations	2,100	2,100	2,100	2,100

Robust standard errors in parentheses
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 4: Estimation of specification (2)

	Trade (1)	Intermediate goods (2)	Consumption goods (3)	Capital goods (4)
$RTA_{ij,t}$	0.094***	0.109***	0.080**	−0.008
	(0.023)	(0.029)	(0.032)	(0.024)
Observations	551,471	548,040	548,433	529,141
Number of pairs	36,633	35,703	35,796	32,818
Exporter × Year FE	Yes	Yes	Yes	Yes
Importer × Year FE	Yes	Yes	Yes	Yes
Pair fixed effects	Yes	Yes	Yes	Yes
Pseudo R^2	0.992	0.990	0.989	0.990

Robust standard errors in parentheses.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 5: Estimation of specification (3)

	Trade	Capital	Intermediate	Consumption
$\ln GDP_i$	1.131***	1.629***	1.196***	0.687***
	(0.000)	(0.000)	(0.000)	(0.001)
$\ln EPU_i$	0.073	-0.163	0.175	0.088
	(0.877)	(0.807)	(0.698)	(0.843)
$\ln EPU_{i,t-2}$	-0.318	-0.373	-0.227	-0.382*
	(0.161)	(0.308)	(0.302)	(0.059)
$\ln EPU_{i,t-4}$	-0.411**	-0.704**	-0.289	-0.385*
	(0.038)	(0.034)	(0.122)	(0.071)
Remoteness	0.246	-0.817	0.915**	-0.919*
	(0.559)	(0.355)	(0.035)	(0.074)
WTO membership	-0.085	1.485***	-0.585**	0.219
	(0.686)	(0.000)	(0.011)	(0.294)
$\ln Population_i$	-0.386***	-0.487*	-0.489***	-0.098
	(0.005)	(0.051)	(0.001)	(0.494)
Observations	273	273	273	273
Countries	21	21	21	21
R^2	0.569	0.579	0.597	0.446

Notes: Robust p-values in parentheses. Year fixed effects included in all specifications but not reported. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 6: Estimation of specification (4)

	Trade	Capital	Intermediate	Consumption
$\ln GDP_j$	0.795***	0.882***	0.748***	0.691***
	(0.000)	(0.000)	(0.000)	(0.000)
$\ln EPU_j$	0.075	0.197	0.032	0.055
	(0.644)	(0.274)	(0.858)	(0.696)
$\ln EPU_{j,t-2}$	-0.180*	-0.160	-0.202*	-0.170**
	(0.058)	(0.152)	(0.050)	(0.029)
$\ln EPU_{j,t-4}$	-0.135	-0.150	-0.129	-0.142
	(0.138)	(0.158)	(0.207)	(0.130)
Remoteness	-0.468	-0.339	-0.529	-0.454
	(0.124)	(0.246)	(0.141)	(0.114)
WTO membership	0.118	-0.266**	0.356***	-0.119
	(0.211)	(0.016)	(0.006)	(0.409)
$\ln Population_j$	-0.138	-0.243***	-0.058	-0.273**
	(0.102)	(0.008)	(0.574)	(0.022)
Observations	273	267	273	272
Countries	21	21	21	21
R^2	0.822	0.804	0.780	0.713

Notes: Robust p-values in parentheses. Year fixed effects included in all specifications but not reported. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.